

$^2\text{H}(^{34}\text{Si}, ^{35}\text{P}\gamma)$     [2007GeZX](#)

$^{34}\text{Si}(\text{d}, \text{n})^{35}\text{P}$  from  $J^\pi=0^+$   $^{34}\text{Si}$  g.s. in inverse kinematics.

[2007GeZX](#): 30-AMeV  $^{34}\text{Si}$  beam on 30-mg/cm<sup>2</sup> CD<sub>2</sub> secondary target at GANIL. Heavy ions produced in reactions were identified by the VAMOS spectrometer.  $\gamma$  rays were detected using the EXOGAM germanium clover array. Measured Doppler-corrected  $E_\gamma$ ,  $I_\gamma$ ,  $\gamma\gamma$ -coin, and  $(^{35}\text{P})\gamma$ -coin. Deduced levels,  $J$ ,  $\pi$ . Compared with shell-model calculations.

 $^{35}\text{P}$  Levels

$E(\text{level})^\dagger$

0  
2386.5 8  
3859.9 8  
4100.9 13  
4492.9 16  
4868.9 13

<sup>†</sup> From a least-squares fit to  $\gamma$ -ray energies, assuming  $\Delta E_\gamma=1$  keV since uncertainties were not given in [2007GeZX](#).

 $\gamma(^{35}\text{P})$ 

| <u><math>E_\gamma^\dagger</math></u> | <u><math>E_i(\text{level})</math></u> | <u><math>E_f</math></u> |
|--------------------------------------|---------------------------------------|-------------------------|
| 241                                  | 4100.9                                | 3859.9                  |
| 392                                  | 4492.9                                | 4100.9                  |
| 1009                                 | 4868.9                                | 3859.9                  |
| 1473 <sup>‡</sup>                    | 3859.9                                | 2386.5                  |
| 2386                                 | 2386.5                                | 0                       |
| 3860                                 | 3859.9                                | 0                       |

<sup>†</sup> From [2007GeZX](#).

<sup>‡</sup> Placement of transition in the level scheme is uncertain.

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Legend

Level Scheme

 - - - - -  $\rightarrow$   $\gamma$  Decay (Uncertain)
